

Tech & Teach

Digital Chip Design Program

RTL Design

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Structural modeling & testbench

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Task #4.1

```

module top (
  input [3:0] A,B,
  input C,
  output Y);
  wire AeqB;
  wire AltB;
  wire AgtB;
  wire n1;
  wire n2;
  comparator U1 (
    .AeqB(AeqB),
    .AltB(AltB),
    .AgtB(AgtB),
    .A(A),
    .B(B));
  Or2 U2 (
    .A(AgtB),
    .B(AltB),
    .Y(n1));
  And2 U3 (
    .A(C),
    .B(C),
    .Y(n2));
  Not1 U4 (
    .A(n2),
    .Y(Y));
endmodule

```

```

module Or2(
  input logic A,B,
  output logic Y);
  assign Y= A|B;
endmodule

```

```

module And2(
  input logic A,B,
  output logic Y);
  assign Y= A&B;
endmodule

```

```

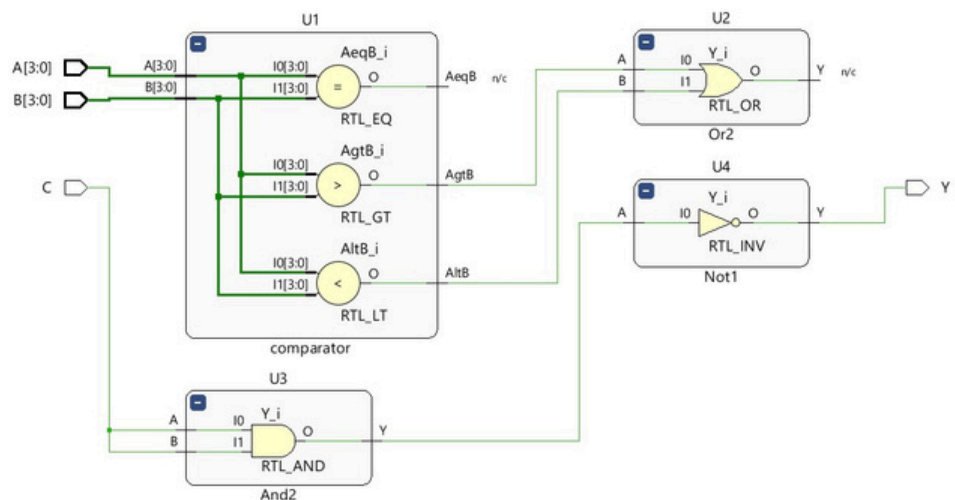
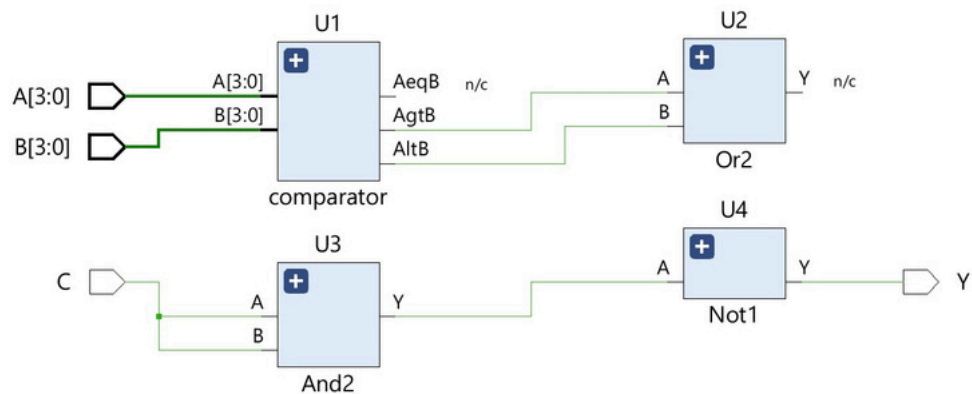
module Not1(
  input logic A,
  output logic Y);
  assign Y= ~A;
endmodule

```

```

module comparator(AeqB, AltB, AgtB, A, B);
  input [3:0] A,B;
  output AeqB, AltB, AgtB;
  assign AeqB = (A == B)? 1 : 0;
  assign AltB = (A < B)? 1 : 0;
  assign AgtB = (A > B)? 1 : 0;
endmodule

```



Task #4.2

prime.sv:
module prime(
input logic [2:0] A,
output logic isPrime
);
assign isPrime = (A[2])? A[0]: A[1];
endmodule

`timescale 1ns/1ps

module prime_tb;

logic [2:0] A;
logic isPrime;

prime dut (
 .A(A),
 .isPrime(isPrime)
);

initial begin
 A = 3'b000; #10;
 A = 3'b001; #10;
 A = 3'b010; #10;
 A = 3'b011; #10;
 A = 3'b100; #10;
 A = 3'b101; #10;
 A = 3'b110; #10;
 A = 3'b111; #10;
 \$finish;
end

endmodule

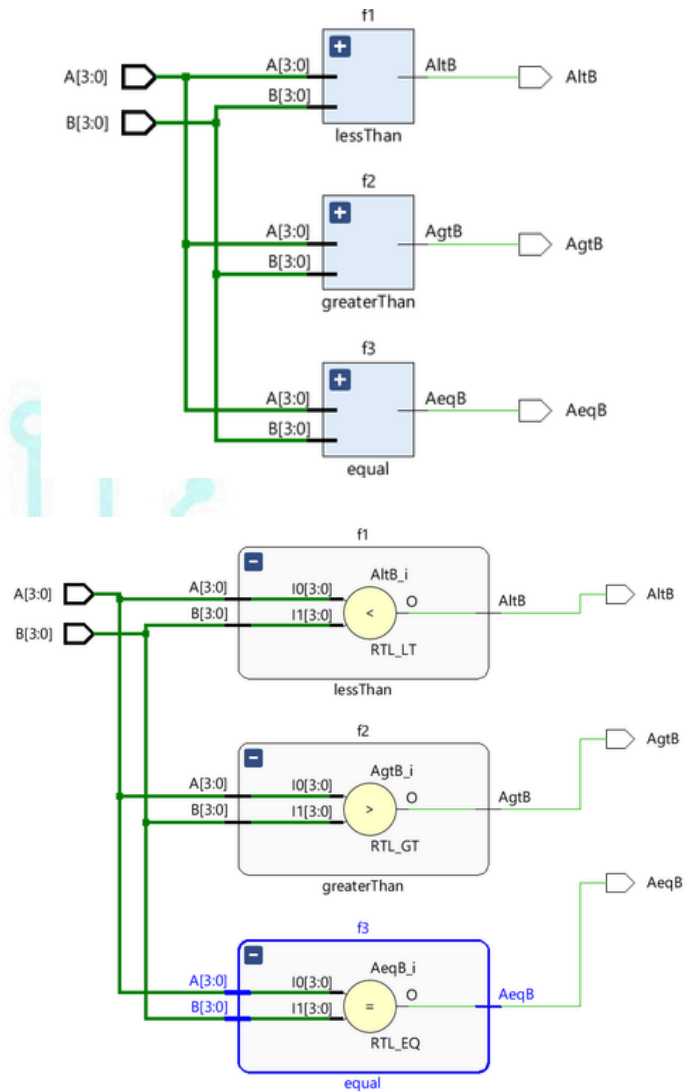
Name	Value
A[2:0]	7
isPrime	1

Task #4.3

```

module top(
input logic [3:0] A,B,
output logic AltB,AgtB,AeqB);
lessThan f1 (
.AltB(AltB),
.A(A),
.B(B));
greaterThan f2 (
.AgtB(AgtB),
.A(A),
.B(B));
equal f3 (
.AeqB(AeqB),
.A(A),
.B(B));
endmodule

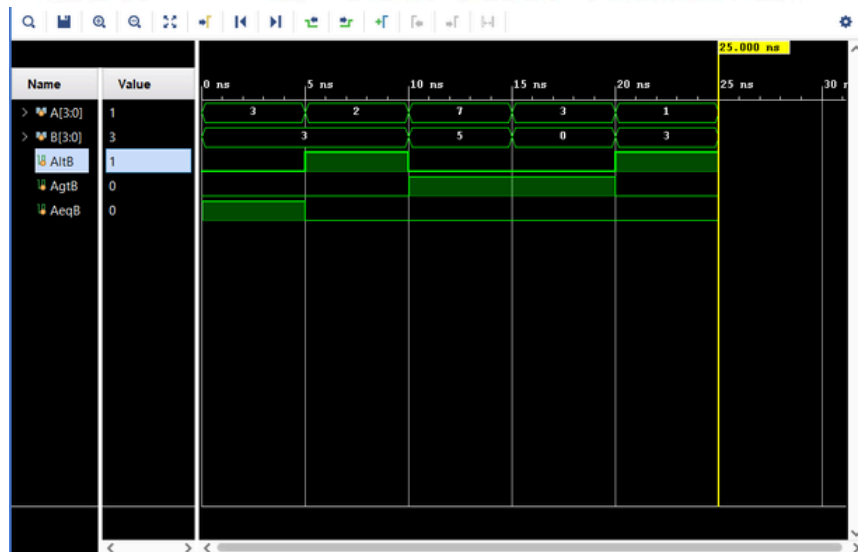
```



```

`timescale 1ns / 1ps
module testbench();
logic [3:0] A,B;
logic AltB,AgtB,AeqB;
top dut (.A(A),
.B(B),
.AltB(AltB),
.AgtB(AgtB),
.AeqB(AeqB));
initial begin
A=4'b0011;B=4'b0011; #5;
A=4'b0010;B=4'b0011; #5;
A=4'b0111;B=4'b0101; #5;
A=4'b0011;B=4'b0000; #5;
A=4'b0001;B=4'b0011; #5;
$finish;
end
endmodule

```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task4.1	✓	✓	✓	✓	✓
Task4.2	✓	✓	✓	✓	✓
Task4.3	✓	✓	✓	✓	✓

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